Prime-Gap Types, Crossed Rescues, and the Limits of a Multitype Induction for Goldbach Representations

Exact Transition Identities, Singular-Series Structure, and Computational Evidence

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Abstract

We investigate an inductive approach to binary Goldbach representations that augments existence with *types* determined by prime-pair gaps. A fixed pair of gaps $(6k+2, 6k+4)$ provides an exact two-endpoint replacement between the consecutive even integers $6r+2$ and $6r+4$, and four associated primes produce a family of four additive targets. We derive a mixed convolution for each rescue type, a sum-and-difference identity in signed imbalance coordinates, and an exact identity for the total number of oriented rescue configurations. We establish local admissibility and a uniform finite-sieve lower bound simultaneously for linearly many types, as well as explicit cross-type modular exclusions. For general even prime gaps (a, b), we give an orientation-aware generating-function formalism and a uniformly positive Euler-product singular series. Computations for thirteen gap types through $r = 300{,}000$ and a fully enumerated example at the complete reset $277 \to 278$ illustrate redundancy among rescue mechanisms. The multitype renewal implication needed for an induction is isolated as a separate open conjecture. None of the results proves Goldbach or any prescribed-gap prime-pair conjecture.

Keywords. Goldbach conjecture; prime pairs; twin primes; cousin primes; sieve methods; singular series; additive convolution; induction; arithmetic dynamics.
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1 Introduction and scope

The binary Goldbach conjecture asserts that every even integer $N \geq 4$ is a sum of two primes. The classical heuristic of Hardy and Littlewood [3] predicts the density of representations but has not yielded a universal proof. We study a more structured question: can a representation at one even target *force* a representation at the next target because the prime endpoints belong to specified prime-pair classes?

The motivating example is the complete reset between $554 = 2 \cdot 277$ and $556 = 2 \cdot 278$:

$$239 + 313 = 552, \quad 241 + 313 = 554, \quad 239 + 317 = 556, \quad 241 + 317 = 558. \tag{1}$$

Here $(239, 241)$ is a twin-prime pair and $(313, 317)$ a cousin-prime pair. The transition from 554 to 556 replaces *both* endpoints. We generalize this replacement rule to a hierarchy of fixed-gap types.

We carefully separate three levels of assertion: identities following from the definitions; local congruence and finite-sieve theorems, which do not require simultaneous primality; and existence or asymptotic conjectures about four actual primes. In particular, the observed persistence of a finite family of rescue types does not supply a valid inductive step without an additional uniform renewal theorem. This distinction is consistent with the obstacles familiar from sieve theory [2, 4].

2 Goldbach rows, resets, and four-prime tiles

For integer $m \geq 3$, define the ordered endpoint set

$$S(m) = \{p \in \mathbb{P} : 2m - p \in \mathbb{P}\}. \tag{2}$$

The set is reflection-invariant: $p \in S(m)$ if and only if $2m - p \in S(m)$. A *complete reset* at $m \to m+1$ means

$$S(m) \cap S(m+1) = \varnothing, \tag{3}$$

which does *not* mean that either row is empty.

Proposition 2.1 (Exact complete-reset criterion)**.** *For every integer $r \geq 2$,*

$$S(3r+1) \cap S(3r+2) = \varnothing \quad \Longleftrightarrow \quad 6r - 1 \notin \mathbb{P}. \tag{4}$$

Proof. A common endpoint p gives primes $q = 6r + 2 - p$ and $q + 2 = 6r + 4 - p$. If $q > 3$, the lower member q of a twin pair satisfies $q \equiv -1 \pmod 6$, apart from no other exception, and hence $p \equiv 3 \pmod 6$, forcing $p = 3$. This gives $q = 6r - 1 \in \mathbb{P}$. If $q = 3$, then $p = 6r - 1 \in \mathbb{P}$. Conversely, if $6r - 1$ is prime, then $p = 6r - 1$ belongs to both rows via complements 3 and 5. □

Definition 2.2 (Oriented four-prime tile)**.** For $k \geq 0$ and $i, j \geq 1$, a *type-k tile* consists of the four primes

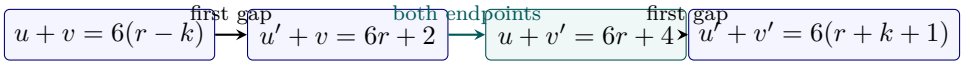
$$u = 6i - 1, \quad u' = 6(i + k) + 1, \quad v = 6j + 1, \quad v' = 6(j + k) + 5. \tag{5}$$

The first gap is $u' - u = 6k + 2$ and the second is $v' - v = 6k + 4$.

Writing $r = i + j + k$, the four sums satisfy

$$\begin{aligned} u + v &= 6(r - k), \quad & u' + v &= 6r + 2, \\ u + v' &= 6r + 4, \quad & u' + v' &= 6(r + k + 1). \end{aligned} \tag{6}$$

Therefore $u' + v \mapsto u + v'$ is a forced two-endpoint rescue, increasing the target by precisely 2. For $k = 0$, the four targets are four successive even integers. For $k > 0$ they are not four consecutive even integers; the middle two remain consecutive.



Type-k bridge through a complete reset

Figure 1: Four additive targets of a single type-k tile. The middle arrow is a two-endpoint replacement. For $k > 0$ the outer arrows span $6k + 2$, not 2.

3 The multitype rescue convolution

For $k \geq 0$, define the two prime-pair indicators

$$U_k(i) = \mathbf{1}_{\mathbb{P}}(6i-1)\mathbf{1}_{\mathbb{P}}(6(i+k)+1), \qquad V_k(j) = \mathbf{1}_{\mathbb{P}}(6j+1)\mathbf{1}_{\mathbb{P}}(6(j+k)+5). \tag{7}$$

Set

$$R_k(r) = \sum_{i=1}^{r-k-1} U_k(i)V_k(r-k-i), \qquad r \geq k+2, \tag{8}$$

and $R_k(r) = 0$ if $r < k+2$. Each nonzero summand is a type-k rescue from $6r+2$ to $6r+4$.

Proposition 3.1 (Tile-convolution identity)**.** *Let*

$$H_k(s) = \sum_{\substack{i+j=s \\ i,j\geq 1}} U_k(i)V_k(j). \tag{9}$$

Then $R_k(r) = H_k(r-k)$ identically. The corresponding power series satisfy

$$\sum_{r\geq 0} R_k(r)z^r = z^k \left(\sum_{i\geq 1} U_k(i)z^i\right)\left(\sum_{j\geq 1} V_k(j)z^j\right). \tag{10}$$

Proof. Both identities follow by setting $j = r-k-i$ and rearranging the finite coefficient sums. □

The special type $k = 0$ is the lower-twin/lower-cousin convolution, for which

$$R_0(r) = \sum_{i+j=r} \mathbf{1}_{\mathbb{P}}(6i-1)\mathbf{1}_{\mathbb{P}}(6i+1)\mathbf{1}_{\mathbb{P}}(6j+1)\mathbf{1}_{\mathbb{P}}(6j+5). \tag{11}$$

Its conjectured universal positivity is *stronger* than Goldbach and implies infinitely many twin and cousin pairs.

Proposition 3.2 (Sufficient rescue condition)**.** *For any $K \geq 0$, if $\sum_{k=0}^{K} R_k(r) > 0$, then $S(3r+2) \neq \varnothing$. If $6r-1$ is composite, this guarantees a nonempty row immediately after a complete reset.*

Proof. Any positive summand supplies the prime pair $u + v' = 6r+4$ in (6). The reset assertion follows from Proposition 2.1. □

4 Imbalance geometry and a conservation identity

To understand how many distinct configurations are being counted, let $m = 3r+1$. Write each *nonexceptional* unordered Goldbach representation of $2m$ as

$$(m-d) + (m+d) = 2m, \tag{12}$$

with $d \geq 0$ and both primes strictly larger than 3. Let D_r denote the set of resulting half-gaps. Likewise, let E_r comprise the half-gaps $e \geq 0$ of nonexceptional representations

$$(m+1-e) + (m+1+e) = 2(m+1). \tag{13}$$

Write $G = |D_r|$, $G' = |E_r|$, $\Delta = \mathbf{1}_{\{0\in D_r\}}$, and $\Delta' = \mathbf{1}_{\{0\in E_r\}}$.

For such nonexceptional pairs the two old endpoints are 1 (mod 6), while the two new endpoints are 5 (mod 6). Consequently $d+e$ and $|d-e|$ are positive and congruent to 3 (mod 6). The old and new midpoints cannot both be prime, so $\Delta\Delta' = 0$.

Theorem 4.1 (Sum-and-difference rescue identity)**.** *For $r \geq 2$ and every $k \geq 0$,*

$$\boxed{R_k(r) = \sum_{d \in D_r} \sum_{e \in E_r} \mathbf{1}_{\{d+e=6k+3\}} + \sum_{d \in D_r \setminus \{0\}} \sum_{e \in E_r \setminus \{0\}} \mathbf{1}_{\{|d-e|=6k+3\}}.} \tag{14}$$

In particular,

$$\boxed{\sum_{k \geq 0} R_k(r) = 2GG' - \Delta G' - \Delta' G.} \tag{15}$$

Proof. Choose an old pair $(m-d, m+d)$ and a new pair $(m+1-e, m+1+e)$. Placing the old larger prime above the new smaller prime gives a downward endpoint displacement $d+e-1 = 6k+2$ and hence one oriented type-k rectangle. If both d, e are nonzero, the comparison of the old and new upper (if $d > e$), or of their lower (if $e > d$), provides a second downward displacement $|d-e|-1 = 6k+2$. These are the only positive displacements of the required residue class, and they correspond bijectively to the configurations in (8). This proves (14). On summing over k, every pair (d, e) contributes once through $d+e$, and contributes once more through $|d-e|$ precisely when $d, e > 0$. Hence $GG' + (G-\Delta)(G'-\Delta') = 2GG' - \Delta G' - \Delta' G$, using $\Delta\Delta' = 0$. □

Remark 4.2 (Why this does not close an induction). The right-hand side of (15) contains the representation count G' of the *new* Goldbach row. If that row is empty, the entire rescue vector vanishes. Thus the conservation identity describes connections between two existing populations; it cannot independently prove that the second population exists. Also, representations involving the prime 3 are intentionally excluded from D_r, E_r; they must not be inserted into (15) without correction terms.

5 Local sieve and correlations among rescue types

For a fixed type k the required four forms are

$$\begin{aligned} L_{1,k}(i) &= 6i - 1, & L_{2,k}(i) &= 6(i+k) + 1, \\ L_{3,k}(i) &= 6(r-k-i) + 1, & L_{4,k}(i) &= 6(r-i) + 5. \end{aligned} \tag{16}$$

Their product is denoted by $Q_{k,r}(i)$. For primes $\ell \geq 5$, the forbidden residues of $6i \pmod{\ell}$ are

$$F_{\ell,k}(r) = \{1, -6k-1, 6(r-k)+1, 6r+5\} \pmod{\ell}. \tag{17}$$

The set has cardinality $\omega_{\ell,k}(r) \leq 4$; thus every individual type is locally admissible modulo every prime $\ell \geq 5$. All forms are odd and are nonzero modulo 3 for all i.

Proposition 5.1 (Exceptional factors of a type)**.** *For $\ell \geq 5$, $\omega_{\ell,k}(r) < 4$ only if ℓ divides at least one of*

$$6k+2, \quad 6k+4, \quad 6(r-k), \quad 6r+2, \quad 6r+4, \quad 6(r+k+1). \tag{18}$$

Consequently the corresponding four-form singular series has a factorization

$$\mathfrak{S}_k(r) = 81 C_0 \prod_{\substack{\ell \geq 5 \\ \ell \mid D_k(r)}} \frac{\ell - \omega_{\ell,k}(r)}{\ell - 4}, \qquad C_0 = \prod_{\ell \geq 5} \frac{1 - 4/\ell}{(1-1/\ell)^4} > 0, \tag{19}$$

where $D_k(r)$ is the product of the six integers in (18), and factors for primes not dividing $D_k(r)$ are 1. In particular $\mathfrak{S}_k(r) \geq 81 C_0$.

Proof. The six pairwise differences of the four entries in (17), up to sign, are exactly (18). At primes not dividing their product, the four residues are distinct and the generic local factor is $(1-4/\ell)/(1-1/\ell)^4$. At an exceptional prime the correction ratio is at least 1. Modulo 2, all forms are odd for every i, so the local factor is 16; modulo 3, none vanishes, giving $(1-1/3)^{-4} = 81/16$. Their product is 81. □

Proposition 5.2 (A cross-type exclusion)**.** *If $r \equiv 0 \pmod 5$ and the four required prime forms for types 1 and 2 all exceed 5, then types 1 and 2 cannot share the same starting index i.*

Proof. Equation (17) gives $F_{5,1}(r) = \{0,1,3\}$ and $F_{5,2}(r) = \{0,1,2,4\}$. Their union is $\mathbb{Z}/5\mathbb{Z}$. Thus for any i, some required form in one of the two types is divisible by 5. Since it exceeds 5, it is composite. □

Mutual exclusion at a *shared starting index* does not imply that either type has a genuine prime witness. Nevertheless, uniform CRT counts show that both types have many locally admissible indices.

Theorem 5.3 (Uniform finite-sieve survival across many types)**.** *Fix $0 < \theta < 1$ and $0 < \varepsilon < 1$. For $0 \le k \le \theta r$, let*

$$\mathcal{C}_{r,k} = \{i \in \mathbb{Z} : \lceil (r-k)/3 \rceil \le i \le \lfloor 2(r-k)/3 \rfloor\}, \tag{20}$$

and let $W_{k,y}(r)$ count $i \in \mathcal{C}_{r,k}$ for which no prime $5 \le \ell \le y$ divides $Q_{k,r}(i)$. There exist $r_0(\theta,\varepsilon)$ and $c_{\theta,\varepsilon} > 0$ such that for every $r \ge r_0$ and every $0 \le k \le \theta r$,

$$\boxed{W_{k,(1-\varepsilon)\log r}(r) \ge c_{\theta,\varepsilon} \frac{r}{(\log\log r)^4}.} \tag{21}$$

Proof. Set $M_y = \prod_{5 \le \ell \le y} \ell$ over primes. By CRT, the allowed proportion of classes modulo M_y is $\rho_{k,y}(r) = \prod_{5 \le \ell \le y}(1 - \omega_{\ell,k}(r)/\ell)$. Uniformly in r, k this is at least $\prod_{5 \le \ell \le y}(1-4/\ell) \gg (\log y)^{-4}$ by Mertens' estimates [6]. The central interval contains $\gg_\theta r$ integers. Counting full residue periods and one partial period gives

$$W_{k,y}(r) = |\mathcal{C}_{r,k}| \rho_{k,y}(r) + O(M_y). \tag{22}$$

For $y = (1-\varepsilon)\log r$, the prime number theorem gives $M_y = r^{1-\varepsilon+o(1)}$, uniformly independently of k. The error is $o(r/(\log\log r)^4)$, proving the claim. □

Remark 5.4 (Exact gap between sieve survival and primality). Theorem 5.3 is a theorem about avoidance of relatively small divisors; it does not imply a single prime quadruple. A uniform upper bound for elimination by the remaining prime divisors, strong enough to leave one genuine tuple for every r, is exactly the missing arithmetic input. Local admissibility alone cannot defeat the sieve parity obstruction.

6 Orientation-free prime-gap classes and a general conjecture

For any positive even d, let

$$P_d = \{p \in \mathbb{P} : p - d \in \mathbb{P} \text{ or } p + d \in \mathbb{P}\}. \tag{23}$$

For even a, b, let $r_{a,b}(N)$ count $N = p + q$ with $p \in P_a$, $q \in P_b$; for $a = b$, use unordered pairs, including a diagonal pair once.

Conjecture 6.1 (Generalized prime-gap Goldbach covering)**.** For fixed positive even a, b, there exists $N_0(a, b)$ such that $r_{a,b}(N) > 0$ for every even $N \geq N_0(a, b)$.

This conjecture would imply infinitely many prime pairs of *every* prescribed positive even gap, a family of problems at least as strong as the fixed-gap prime-pair conjectures.

It is necessary to distinguish unweighted membership from *oriented* prime-gap multiplicity. Put

$$D_d(z) = \sum_{p,p+d\in\mathbb{P}} z^p, \qquad T_d(z) = \sum_{p-d,p,p+d\in\mathbb{P}} z^p, \tag{24}$$

where p is restricted to positive primes throughout. The endpoint generating function is

$$\boxed{P_d(z) = (1 + z^d)D_d(z) - T_d(z).} \tag{25}$$

The subtraction removes duplicate contributions from primes with prime neighbors at both distances d.

Proposition 6.2 (Exact representation generating functions)**.** *For $a \neq b$, $r_{a,b}(N) = [z^N]P_a(z)P_b(z)$. When $a = b = d$,*

$$r_{d,d}(N) = \frac{1}{2}[z^N](P_d(z)^2 + P_d(z^2)). \tag{26}$$

Proof. The first identity is the ordinary Cauchy product of the indicator series. For equal classes, the ordered coefficient counts each off-diagonal unordered pair twice and each diagonal pair once; adding the diagonal coefficient and dividing by two gives the stated formula. □

Consider, for a fixed orientation $(\eta, \delta) \in \{\pm 1\}^2$, the forms

$$x, \qquad x + \eta a, \qquad N - x, \qquad N - x + \delta b. \tag{27}$$

Let $\nu_\ell^{\eta,\delta}$ denote the number of residues modulo prime ℓ at which all four forms are nonzero. Define the orientation-summed singular series

$$\mathfrak{S}_{a,b}(N) = \chi_{a,b} \sum_{\eta,\delta\in\{\pm1\}} \prod_{\ell} \frac{\nu_\ell^{\eta,\delta}/\ell}{(1 - 1/\ell)^4}, \quad \chi_{a,b} = \begin{cases} 1/2, & a = b, \\ 1, & a \neq b. \end{cases} \tag{28}$$

This counts orientation-weighted configurations; a prime having two partners at gap d can contribute twice. For fixed gaps such multiple orientations are expected to be lower-order in a four-prime asymptotic, but no such asymptotic is proved here.

Theorem 6.3 (Uniform local positivity)**.** *For every positive even a, b, N, at least one orientation is locally admissible at all primes. If $N > a + b$, the Euler products in (28) are well-defined and*

$$\boxed{\mathfrak{S}_{a,b}(N) \geq \frac{27}{4} C_0 > 0.} \tag{29}$$

More precisely, the aggregate number of admissible residues modulo 3, summed over orientations, is

$$\sum_{\eta,\delta} \nu_3^{\eta,\delta} = 2^{\mathbf{1}_{\{3|a\}} + \mathbf{1}_{\{3|b\}} + \mathbf{1}_{\{3|N\}}}. \tag{30}$$

Proof. For $\ell \geq 5$ at most four residues, $0, -\eta a, N, N + \delta b$, are forbidden. Modulo 2, exactly one residue survives because all three parameters are even. Modulo 3, each possible nonzero choice of x and $N - x$ can be extended to a compatible sign of a and b (two signs when the corresponding gap is $0 \pmod 3$, one otherwise). There are two possible x residues if $3 \mid N$, and one otherwise, yielding (30) and local admissibility for at least one orientation. The factors at $\ell = 2, 3$ multiply to $(27/2)\nu_3^{\eta,\delta}$; for $\ell \geq 5$ every collision among forbidden residues increases the local factor relative to C_0. Thus (28) is at least $\chi_{a,b}(27/2)C_0 \sum_{\eta,\delta} \nu_3^{\eta,\delta} \geq (27/4)C_0$. □

For $N > a + b$ the six possible congruence collisions for each orientation are controlled by the nonzero integer

$$D_{\eta,\delta} = abN(N + \eta a)(N + \delta b)(N + \eta a + \delta b). \tag{31}$$

Thus the singular series factors into C_0 times explicit corrections at primes dividing $D_{\eta,\delta}$, and the multiplier from the prime 3 is exactly described by (30). No simple universal ordering by $a = b$ or by divisibility by 3 follows, because other small-prime corrections can reverse the comparison. For example, direct rational calculation gives

$$\mathfrak{S}_{2,4}(104) = \frac{1500}{91}C_0, \qquad \mathfrak{S}_{6,6}(122) = \frac{18271}{475}C_0. \tag{32}$$

The lower bounds $\mathfrak{S}_{10,10}(N) \geq (81/4)C_0$ and $\mathfrak{S}_{2,30}(N) \geq 54C_0$ show that even the minima across N need not be ordered by the proposed residue-3 heuristic.

Conjecture 6.4 (Four-prime Hardy–Littlewood asymptotic)**.** For fixed even $a, b > 0$, as even $N \to \infty$,

$$r_{a,b}(N) \sim \mathfrak{S}_{a,b}(N) \sum_{m=2}^{N-2} \frac{1}{\log^2 m \, \log^2(N - m)} \sim \mathfrak{S}_{a,b}(N) \frac{N}{\log^4 N}. \tag{33}$$

The asymptotic is consistent with Hardy–Littlewood prime-tuple heuristics [1, 3, 5]. If established with the stated relative error, it would imply Conjecture 6.1 by Theorem 6.3. It is not a consequence of local admissibility alone.

7 A complete reset at $277 \to 278$

For $r = 92$ the earlier Goldbach target is $6r + 2 = 554$ and the new target is $6r + 4 = 556$. Since

$$6r - 1 = 551 = 19 \cdot 29, \tag{34}$$

Proposition 2.1 establishes a complete reset.

There are exactly 11 unordered prime pairs summing to 554 and 11 summing to 556, with no prime endpoint 3 in either row. The old row contains the diagonal pair $(277, 277)$ and the new row has no diagonal pair. Thus $(G, G', \Delta, \Delta') = (11, 11, 1, 0)$, and Theorem 4.1 gives

$$\boxed{\sum_{k \geq 0} R_k(92) = 231.} \tag{35}$$

Exactly 72 different k have positive counts; the largest is $k = 83$. The first several counts are:

k	0	1	2	3	4	5	6	7
(a_k, b_k)	(2,4)	(8,10)	(14,16)	(20,22)	(26,28)	(32,34)	(38,40)	(44,46)
$R_k(92)$	1	5	8	6	1	2	7	7

The unique type-0 witness is $(i, j) = (40, 52)$, giving the four primes $239, 241, 313, 317$ and the tile (1). The arithmetic rescue is not the only one: $R_1(92) = 5$, for example, includes the transition $97 + 457 \to 89 + 467$.

Remark 7.1. The 231 oriented rescue configurations do not constitute 231 logically independent witnesses for Goldbach. Formula (15) shows that many configurations are generated by different pairings of the same 11 representations on either side of the reset.

8 Numerical observations for thirteen types

Using an Eratosthenes sieve for primes and discrete convolutions of the integer-valued sequences in (7), we evaluated $R_k(r)$ for $0 \le k \le 12$ and $2 \le r \le 300{,}000$. A floating-point FFT was rounded to the nearest integer; all transforms had residual from integrality below 10^{-4}, and selected indices were checked using direct integer summation. These are reproducible finite observations rather than mathematical proofs of eventual positivity.

Type k	Gap pair (a_k, b_k)	Zeros in $k+2 \le r \le 300000$	Last zero
0	(2,4)	0	—
1	(8,10)	0	—
2	(14,16)	2	25
3	(20,22)	1	5
4	(26,28)	2	27
5	(32,34)	0	—
6	(38,40)	0	—
7	(44,46)	5	30
8	(50,52)	1	10
9	(56,58)	2	27
10	(62,64)	3	167
11	(68,70)	1	13
12	(74,76)	4	105

Table 1: Computed zero counts for thirteen oriented rescue types. Each type is tested only where $r \ge k + 2$.

In particular, the computed data give $R_k(r) > 0$ for *every* $0 \le k \le 12$ and every $168 \le r \le 300{,}000$. At $r = 92$ the first five counts are $(1, 5, 8, 6, 1)$, while at $r = 1000$ they are $(27, 12, 10, 15, 26)$. Type $k = 2$ illustrates non-monotone dynamics:

$$R_2(24) = 2, \qquad R_2(25) = 0, \qquad R_2(26) = 3. \tag{36}$$

This directly falsifies any putative induction asserting that positivity of an individual type necessarily propagates to the next index. The remarkable abundance of *other* types at such steps motivates multitype rather than scalar renewal.

9 What a type induction would need to prove

Define the aggregate rescue count through type K by

$$\mathcal{R}_K(r) = \sum_{k=0}^{K} R_k(r). \tag{37}$$

From Proposition 3.2, $\mathcal{R}_K(r) > 0$ is sufficient for replenishment at the complete reset immediately before $6r + 4$.

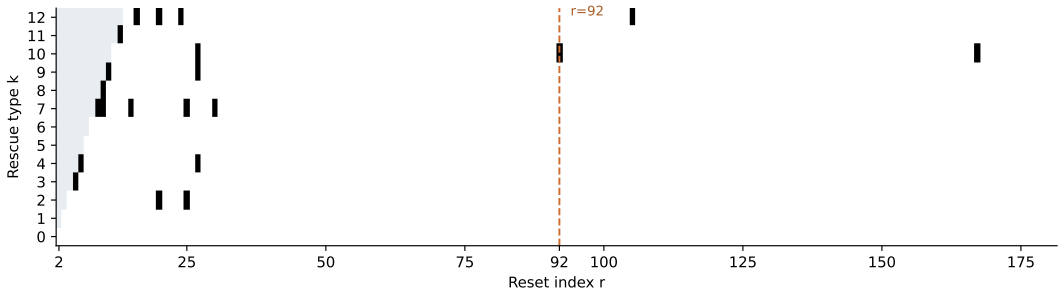


Figure 2: Zeros (dark squares) for the first thirteen types over the first 180 admissible indices, computed from the archived type-count array. The vertical marker indicates $r = 92$.

Conjecture 9.1 (Bounded-type renewal)**.** There are absolute constants K_0 and r_0 such that

$$\boxed{\mathcal{R}_{K_0}(r) > 0 \quad \text{for all } r \geq r_0.} \tag{38}$$

The case $K_0 = 0$ is the lower-twin/lower-cousin convolution conjecture. For arbitrary fixed K_0, (38) remains a strong prime-quadruple covering conjecture; it does not follow from Theorem 5.3.

A putative inductive proof might use a state vector

$$\mathbf{R}(r) = (R_0(r), R_1(r), \ldots, R_{K_0}(r))^{\mathsf{T}} \tag{39}$$

and seek a renewal implication between $\mathbf{R}(r)$ and $\mathbf{R}(r+1)$. There are two genuine algebraic restrictions:

(i) individual prime tiles force all four of their associated sums, including the two consecutive central targets;

(ii) different types share some prime forms or encounter incompatible modular restrictions, as in Proposition 5.2.

Neither restriction yields a lower bound for $\mathcal{R}_{K_0}(r+1)$ from $\mathcal{R}_{K_0}(r)$ alone. The sum identity (15) is also not a renewal law: it explicitly involves the target population one is trying to prove nonempty.

For a precise sieve formulation, choose a finite-sieve threshold y and write

$$\mathcal{R}_K^{\mathrm{cent}}(r) = \sum_{k=0}^{K} W_{k,y}(r) - \mathcal{D}_{K,y}(r), \tag{40}$$

where $\mathcal{D}_{K,y}(r)$ counts, *with type and starting index multiplicity*, the small-prime survivors for which at least one of the four forms is composite. Therefore a sufficient condition is

$$\boxed{\mathcal{D}_{K,y}(r) < \sum_{k=0}^{K} W_{k,y}(r).} \tag{41}$$

Theorem 5.3 bounds the right-hand side for $k \leq \theta r$; no matching uniform upper bound for the left-hand side is known. In particular, local mutual exclusion of two types does not establish that either type contains a prime quadruple.

10 Conclusion

The type formalism adds information absent from the mere indicator of Goldbach representability: it distinguishes endpoint-preserving transitions, two-endpoint replacements, and different gap channels. The resulting convolution and imbalance identities are exact; every type is locally admissible, and linearly many types simultaneously possess abundant small-prime sieve survivors. Cross-type congruence obstructions produce a nontrivial correlation geometry, while the computations show considerable empirical redundancy.

The mathematical bottleneck is clear. The arithmetic of the actual primes must force *new* type populations to appear, rather than merely transport already existing four-prime tiles. Establishing a multitype renewal inequality such as (41) or Conjecture 9.1 would represent a major advance; it cannot be inferred from the current local and algebraic results. We regard the framework as a source of testable conjectures and exact identities, not as a proof of Goldbach.

Computational reproducibility

The source package includes the original FFT-based thirteen-type scanner, its compressed count array, and an independent direct-primality verifier for the $r = 92$ example and the local singular-series corrections. Table 1 and Figure 2 can be reproduced from these materials. No inference about indices beyond the tested range is made.

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